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Version: 1
Issued Date: 2025/07/07

Datasheet

产品名称 (Product): BT 6.0 module

产品型号 (Model No.): HOLYIOT-25007 -nRF54L15

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1. Description

NRF54L15 is the first system level chip (SoC) in the nRF54L series. It is an ultra-low power Bluetooth 5.4 SoC with the best in class new multi protocol radio and advanced security features. The nRF54L series takes the popular nRF52 series to a new level with a more compact packaging, featuring excellent processing power and efficiency, expanded memory, and new peripherals. Main advantages Taking the nRF52 series to the next level: achieving significant leaps in performance, efficiency, and security. Resisting security threats: Advanced security features, including secure boot, secure firmware updates, secure storage, and physical attack protection.Providing long-distance: the best in class multi protocol radio with low power consumption for 1 Mbps Bluetooth, with a transmission power of up to 8 dBm and a reception sensitivity of -98 dBm.Extending battery life or reducing battery size: efficient processing, ultra-low power radio, and minimum sleep current.

Hardware :

SWD programmer (SWDIO,SWCLK,VDD,GND)

nRF54L15 -QFAA-QFN48

Size : 24mm*15.6mm

BLE stack & RF 2.4Ghz

Features :

Excellent processing ability and efficiency

The nRF54L15 SoC integrates an Arm Cortex-M33 processor with a operating frequency of 128 MHz, which has twice the processing power of the nRF52840 and reduces power consumption. It has 1.5 MB of non-volatile memory and 256 KB of RAM, which is sufficient to run multiple wireless protocols simultaneously.

Advanced Security

NRF54L15 is designed specifically for PSA certification level 3, which is the highest level of PSA certified IoT security standard. It provides security services such as secure boot, secure firmware updates, and secure storage. The integrated tampering sensor can detect attacks and take appropriate measures, while the encryption accelerator can be reinforced to resist side channel attacks.

A first-class multi protocol radio

This best in class multi protocol RF product has higher robustness and longer transmission distance. For 1 Mbps Bluetooth with low power consumption, the transmission power can reach 8 dBm (with a step of 1 dB), and the receiving sensitivity is -98 dBm. This radio includes a new 4 Mbps data rate option for 2.4 GHz proprietary protocols, with improved throughput,

efficiency, and latency. It supports low-power Bluetooth and all Bluetooth 5.4 features, and is specifically designed to support the upcoming Bluetooth specifications. In addition, it also supports protocols such as Bluetooth grid, Thread, and Matter.

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New Global RTC peripherals that can extend battery life

The new global RTC peripherals can wake up SoC from the deepest sleep mode, eliminating the need for external RTC, greatly reducing energy consumption for long-term sleep applications and extending battery life.

Ultra compact packaging suitable for size limited designs

NRF54L15 samples are now available in QFN packaging. It will also adopt two ultra compact WLCSP packages. These packages will be 50% smaller than nRF52840 and are suitable for designs with strict size limitations.
6x6 mm QFN48 with 31 GPIOs
2.4x2.2 mm WLCSP with 32 GPIOs (300 um pitch)
2.4x2.2 mm WLCSP with 14 GPIOs (350 um pitch)

Application:

Computer accessories, game controllers, and remote controllers
Virtual reality and augmented reality
Smart Home and Materials
Medical equipment
Industrial Internet of Things

2. Introduction

2.1 Programmer

HOLYIOT-24005- NRF54L15 module use the Serial Wire Debug(SWD port), the module which layout the SWDIO, SWCLK, VDD, GND for debug and flash your own firmware, more info about the SWD, please visit https://www.silabs.com/community/mcu/32-bit/knowledge-base.entry.html/2014/10/21/serial_wire_debugs-qKCT
You can using the Jlink or Jtag for programmer.

2.2 Software development Tool

It supports the standard Nordic Software Development Tool-chain using Segger Embedded Studio, Keil, IAR and GCC. More info please visit https://www.nordicsemi.com/Software-and-Tools/Development-Tools

2.3 Protocols

This module support Bluetooth 5, Bluetooth Low Energy,Bluetooth mesh,Thread,802.15.4,ANT, 2.4GHz proprietary. So we can use different protocols for different situations.

Software Development Kit
Nordic Semiconductor's Software Development Kits (SDK) are your starting point for software development on the nRF51 and nRF52 Series. It contains source code libraries and example applications covering wireless functions, libraries for all peripherals, bootloaders, wired and OTA FW upgrades, RTOS examples, serialization libraries.
More info please visit https://www.nordicsemi.com/Software-and-Tools/Software/nRF5-SDK
You can also download the SDK for coding development .

2.4 SoftDevices

Nordic Semiconductor protocol stacks are known as SoftDevices. SoftDevices are pre-compiled, pre-linked binary files. SoftDevices can be programmed in nRF5 series devices, and are freely downloadable from the Nordic website. Please download that here:

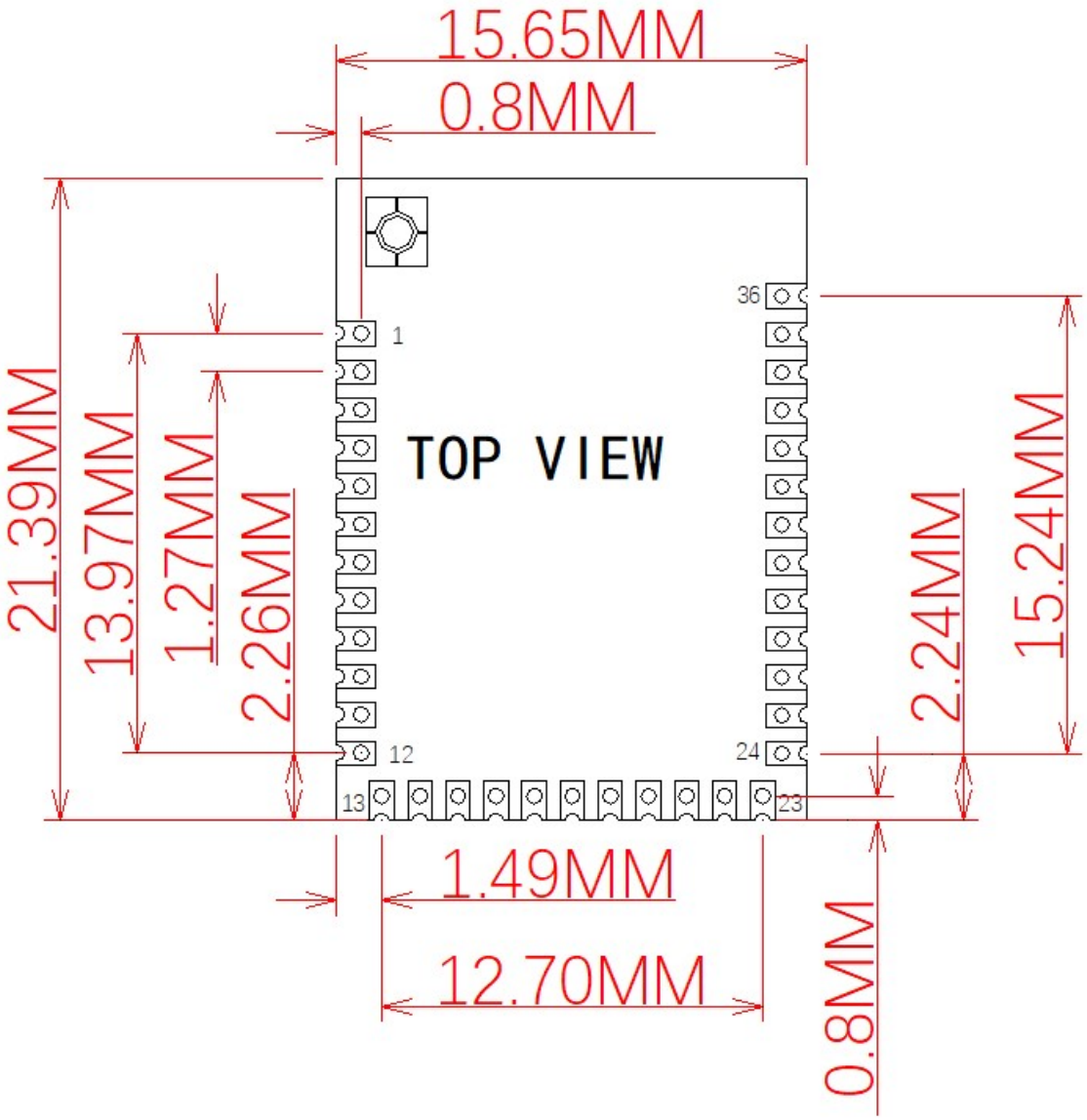
https://www.nordicsemi.com/Software-and-Tools/Software/S132

Over-The-Air DFU

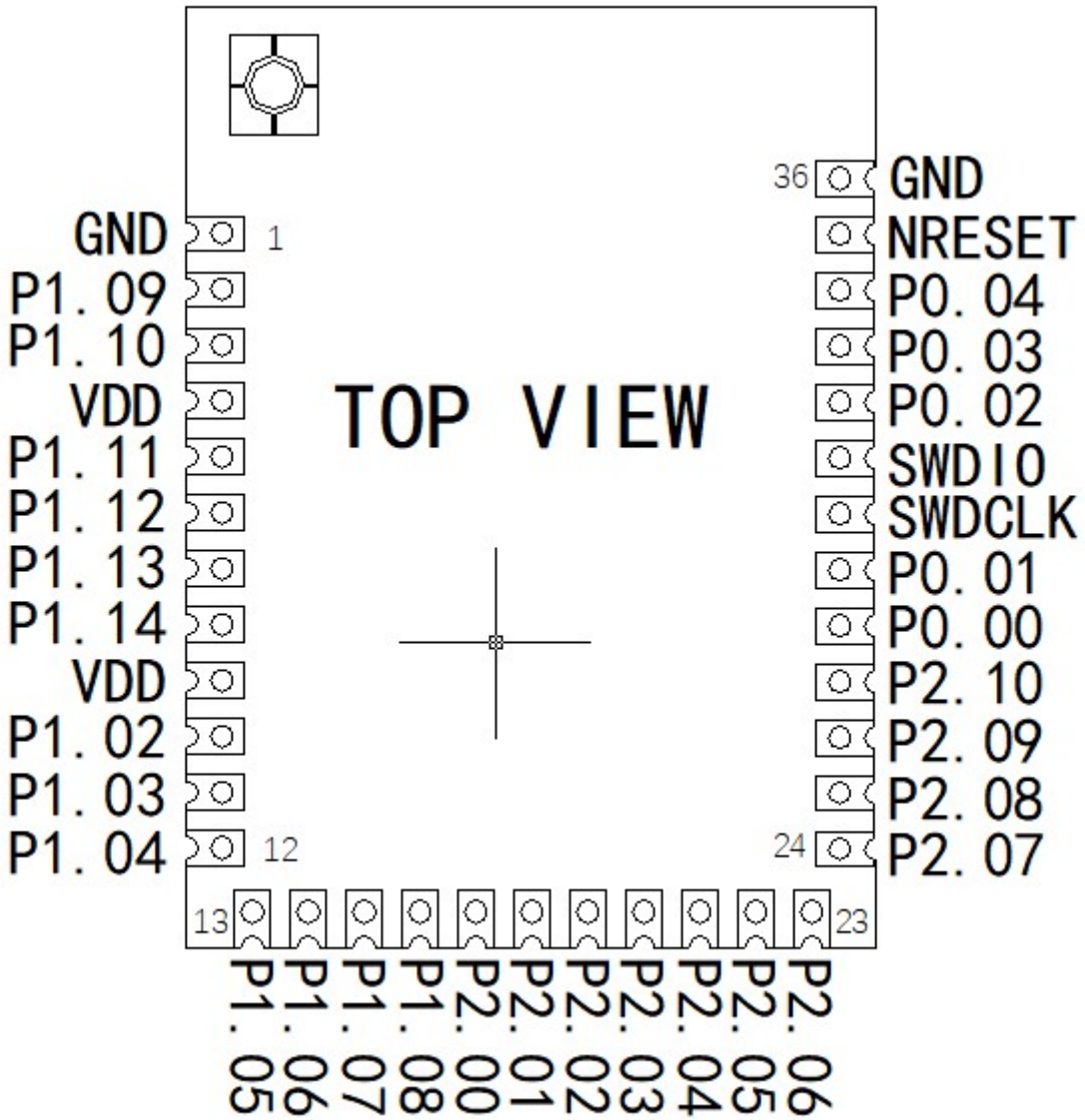
The SoC is supported by an Over-The-Air Device Firmware Upgrade (OTA DFU) feature. This allows for in the field updates of application software and SoftDevice.

3. Product Descriptions

3.1 Mechanical drawings



3.2 Pin assignments



PIN No.	PIN define	Functions
1	GND	Ground
2	P1.09 TAMPC.ASO[2]	Digital I/O Digital I/O
3	P1.10 TAMP.ASI[2]	Digital I/O Digital I/O
4	VDD	power
5	P1.11	Digital I/O

	TAMPC.ASO[3] AIN4	Digital I/O Analog input
6	P1.12 TAMPC.ASI[3] AIN5	Digital I/O Digital I/O Analog input
7	P1.13 AIN6	Digital I/O Analog input
8	P1.14 AIN7	Digital I/O Analog input
9	VDD	power
10	P1.02 NFC1	Digital I/O NFC input
11	P1.03 NFC2	Digital I/O NFC input
12	P1.04 AIN0 ASO[0]	Digital I/O Analog input Digital I/O
13	P1.05 AIN1 ASI[0]	Digital I/O Analog input Digital I/O
14	P1.06 AIN2 ASO[1]	Digital I/O Analog input Digital I/O
15	P1.07 AIN3 ASI[1]	Digital I/O Analog input Digital I/O
16	P1.08 GRTCHFOUT TAMPC.EXT EXTREF	Digital I/O Digital I/O Digital I/O Analog input
17	P2.00	Digital I/O
18	P2.01	Digital I/O
19	P2.02 SWO	Digital I/O Digital I/O
20	P2.03	Digital I/O
21	P2.04	Digital I/O

22	P2.05	Digital I/O
23	P2.06 TRACECLK	Digital I/O Digital I/O
24	P2.07 TRACEDATA[0] SWO	Digital I/O Digital I/O Digital I/O
25	P2.08 TRACEDATA[1]	Digital I/O Digital I/O
26	P2.09 TRACEDATA[2]	Digital I/O Digital I/O
27	P2.10 TRACEDATA[3]	Digital I/O Digital I/O
28	P0.00	Digital I/O
29	P0.01	Digital I/O
30	SWDCLK	Debug
31	SWDIO	Debug
32	P0.02	Digital I/O(general purpose I/O)
33	P0.03 GRTCPWM	Digital I/O(general purpose I/O Trace port clock output)
34	P0.04 GRTCLFCLKOUT	Digital I/O Digital I/O
35	nRESET	Reset
36	GND	Ground

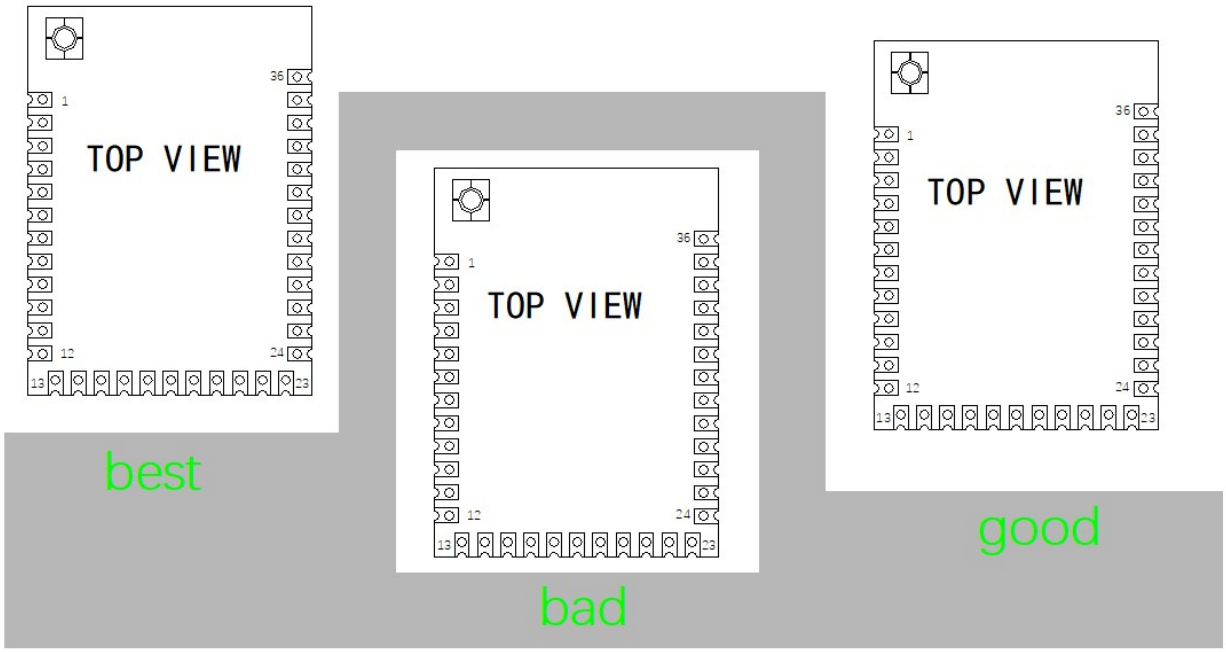
4. Mounting our board on the host PCBA

We suggest that you mount our RF board(HOLYIOT-24005 -nRF54L15) on the board like that:

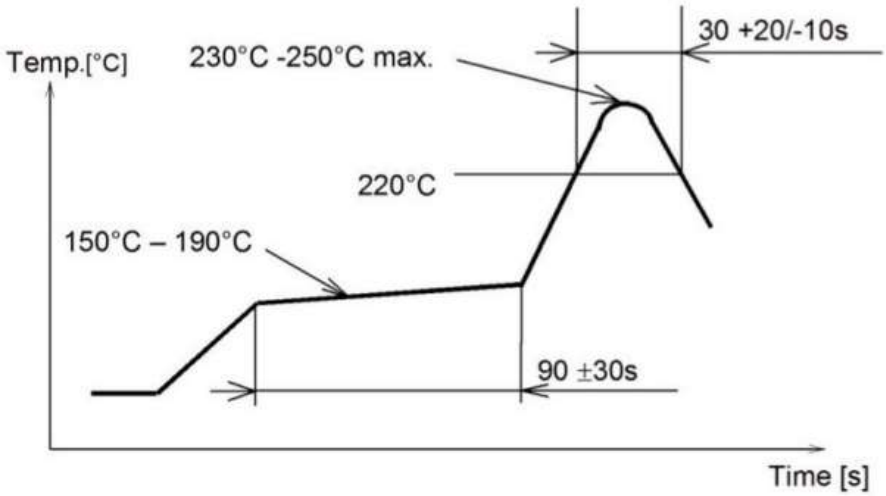
For the best Bluetooth performance, the antenna of the area need to extend about several mm without ground under the antenna of the edge of the host PCB.

The second choice is that place our board at the corner of host PCB, the antenna of board need to extend several mm outside of the Ground plane of the host PCB.

5. Miscellaneous



Soldering Temperature-Time Profile for Re-Flow Soldering. Maximum number of cycles for re-flow is 2. No opposite side re-flow is allowed due to module weight.



6. Recommended operating conditions

Maximum ratings are the extreme limits to which the chip can be exposed for a limited amount of time without permanently damaging it. Exposure to absolute maximum ratings for prolonged periods of time may affect the reliability of the device.

	Parameter	Min.	Max.	Unit
VDD	VDD supply voltage	-0.3	3.9	V
VDD_{EXT}	VDD supply voltage under extended operating temperature	-0.3	3.7	V

Table 85: Supply voltage

	Parameter	Min.	Max.	Unit
$V_{I/O}$, VDD ≤3.6 V	IO voltage	-0.3	VDD + 0.3	V
$V_{I/O}$, VDD >3.6 V	IO voltage	-0.3	3.9	V
$V_{I/O,EXT}$, VDD_{EXT} ≤3.4 V	IO voltage under extended operating temperature	-0.3	VDD + 0.3	V
$V_{I/O,EXT}$, VDD_{EXT} >3.4 V	IO voltage under extended operating temperature	-0.3	3.7	V

Table 86: I/O pin voltage

	Min.	Max.	Unit
RF input level		10	dBm

Table 87: Radio

	Note	Min.	Max.	Unit
Storage temperature		-40	+125	°C
Reflow soldering temperature	Reflow cycle time is 30 seconds with 3 maximum reflow cycles.		260	°C
Moisture Sensitivity Level (MSL)			2	
ESD Human Body Model (HBM)			1	kV
ESD Charged Device Model (CDM)			500	V

	Note	Min.	Max.	Unit
Storage temperature	Recommended storage condition is < 40°C and < 90% RH (relative humidity)	-40	+125	°C
Reflow soldering temperature	Reflow cycle time is 30 seconds with 3 maximum reflow cycles.		260	°C
Moisture Sensitivity Level (MSL)			1	
ESD Human Body Model (HBM)			1	kV
ESD Charged Device Model (CDM)			250	V

Table 89: Environmental CSP package types

	Min.	Max.	Unit
Endurance	10,000		Write/ rewrite cycles
Retention at 85°C	10		y
Retention at 105°C	2		y

Table 90: RRAM memory

